Data Analytics Tasks & Instructions

- **TASK 1: Professional LinkedIn Announcement**
  - **Objective:** Share your official internship acceptance on your professional network.
  - **Requirements:** Post a high-resolution copy of your official **Internship Offer Letter** on LinkedIn. Craft a compelling caption detailing your learning goals, tag **Progree**, and use hashtags: `#dataanalytics #python #tableau #progree`

- **TASK 2: Exploratory Data Analysis & Integrity Cleaning Pipeline**
  - **Objective:** Extract descriptive insights and process uncurated data sources.
  - **Requirements:** Load a large raw industry dataset into an analytics environment (Python/Pandas or R). Diagnose data abnormalities, execute outlier filtering treatments, impute missing parameters systematically, and plot descriptive variable distributions to identify key patterns.

- **TASK 3: Customer Segmentation & Behavioral Trend Dashboard**
  - **Objective:** Design an interactive data portal to isolate distinct consumer behaviors.
  - **Requirements:** Compute complex customer metrics (e.g., Recency, Frequency, Monetary data points). Export the clean data structures into a business intelligence suite (Tableau or Power BI) to generate interactive charts, cohort trends, and geospatial filter tracking maps.

- **TASK 4: Mini Project - Statistical Inferential Hypothesis Analysis & Predictive Business Forecaster**
  - **Objective:** Perform complex hypothesis tests and build predictive trend regressions on enterprise metrics.
  - **Requirements:** Formulate and execute formal statistical significance tests (e.g., A/B test validations, ANOVA mappings, Chi-Square evaluations) to prove experimental corporate outcomes. Build and evaluate multivariate regression or time-series forecasting models (e.g., ARIMA/SARIMA) to project demand variables, complete with detailed confidence performance boundaries.

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